
Q/R/M/C: Stage Decomposition for Memory-Based Navigation in Single Agents and Cadence-Coupled Collectives

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Abstract

Success-only metrics collapse qualitatively different memory-based navigation failures into a single scalar, hiding which stage of the retrieval pipeline is responsible. The Q/R/M/C decomposition — query formation, retrieval satisfaction, target materialization, completion after retrieval — is proposed and proved identifiable from trajectory logs under a conditional stage-Markov assumption (Theorem 1).

Single-agent. On a 10-seed MiniGrid benchmark the framework localizes bottlenecks: hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$) while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without moving end-to-end success).

Multi-agent. On a 35,640-run sweep with broadcast cadence $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$ and 40 seeds, the bottleneck-shift signature appears as a strongly negative within-configuration correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ (Pearson -0.61 at $K = 4$, decaying to -0.02 at $K = 64$), and mean time-to-success has an interior minimum at $K = 8$.

Honest limit. The strict interior-maximum corollary on the conditional product $P(M^*|R^*) \cdot P(C^*|M^*)$ does *not* hold — the product monotonically approaches the independent-agent boundary. This is reported as a feature of the framework, not a bug: the slope-level claim of Proposition 2 is supported while one of its corollaries is not.

1 Introduction

Classical navigation benchmarks often assume that the agent is given an explicit destination coordinate or a dense task reward. Many practical navigation problems are different: the agent must reach a place of a certain *type* — a safe rest area, a water-like resource, a post-hazard recovery exit, or a goal-associated region — specified by a semantic pattern rather than a fixed (x, y) waypoint. This setting opens a methodological question: can we measure *where* memory-based semantic navigation breaks down, not only *whether* it succeeds?

We propose a four-stage diagnostic framework that answers this question for both single agents and multi-agent collectives. The decomposition is Q/R/M/C: **Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, and **C**ompletion after retrieval. The decomposition factorizes per-episode success and, under a conditional stage-Markov property, is identifiable from trajectory logs (Theorem 1).

Single-agent diagnostics. Success-only metrics collapse qualitatively different failures into a single scalar. In a 10-seed MiniGrid benchmark, water and rest tasks are strong end-to-end (0.95, 0.93 success), but goal-region rejoin and hazard recovery are substantially harder. The four-stage view turns this from a uniform weakness into actionable attribution: hazard recovery is retrieval-limited

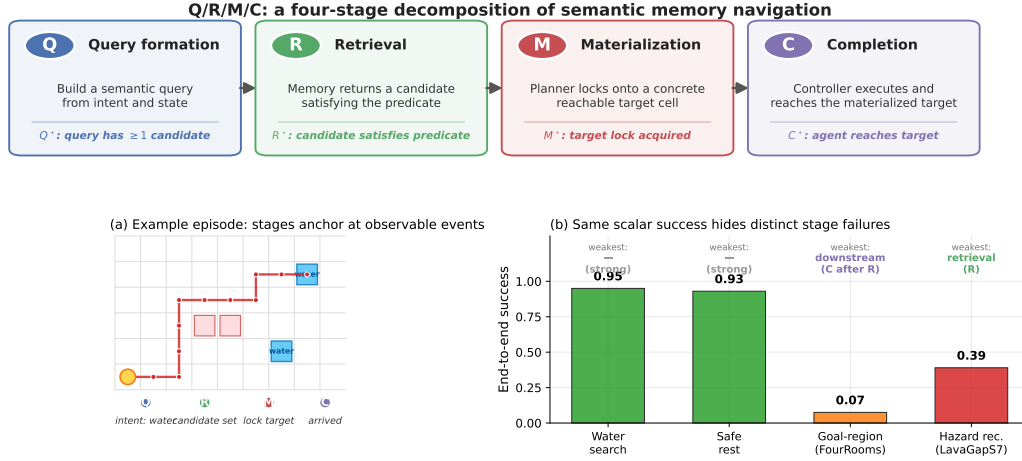


Figure 1: Q/R/M/C decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events Q^* , R^* , M^* , C^* . **(b)** The same scalar “end-to-end success” summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §7.4).

(oracle retrieval lifts success $0.39 \rightarrow 0.60$), while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without changing end-to-end success). The framework therefore produces stage-specific prescriptions instead of undifferentiated evaluations.

Multi-agent extension. When N agents share memory through a communication protocol with broadcast cadence K , the single-agent factorization extends with enlarged conditioning sets (Proposition 1^{*}). The cadence K becomes a structural axis that the framework allows investigation of diagnostically rather than end-to-end.

Bottleneck shift between M and C stages. A slope-level claim (Proposition 2) is stated under stylized assumptions on freshness and occupancy: as K decreases, $P(M^*|R^*)$ rises (fresher peer memory enables reliable commitment) and $P(C^*|M^*)$ falls (peers race for the same targets). Both slope signs are confirmed on the 35,640-run sweep ($p < 10^{-4}$).

What the data do and do not support. A naive reading of the two slopes predicts an interior maximum on the conditional product $P(M^*|R^*) \cdot P(C^*|M^*)$. In this environment that prediction fails: the product is monotone in K and approaches the independent-agent boundary. The interior optimum the framework intuitively predicts appears instead on mean time-to-success, with a clean minimum at $K = 8$. The gap is reported explicitly — it is the difference between a stage-level operationalisation that can be *wrong* on specific summaries and a stage-agnostic guess that is unfalsifiable.

Why this framing is timely. Embodied and goal-conditioned agents increasingly retrieve places, objects, or tools by meaning (Andrychowicz et al., 2017; Ghosh et al., 2021; Savva et al., 2019). LLM-agent systems lean on explicit memory layers but evaluate mostly end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Multi-agent reinforcement learning likewise increasingly involves communicating populations (Lowe et al., 2017; Foerster et al., 2018), but the question “which agent should know what, and when?” lacks a unified diagnostic vocabulary. Our framework supplies one.

Contributions.

- Theorem.** A stage-decomposition framework with a probabilistic factorization and an identifiability result for single agents (Theorem 1) and its multi-agent extension (Proposition 1^{*}).
- Slope-level claim.** A bottleneck-shift claim (Proposition 2) about how cadence K trades off the M and C stages, verified empirically ($p < 10^{-4}$ on both slopes).

- 65 3. **System.** A symbolic-semantic memory stack instantiating the framework, with four communi-
66 cation architectures (independent, shared-bus, central-aggregator, peer-to-peer broadcast).
- 67 4. **Benchmarks.** (a) 10-seed single-agent MiniGrid benchmark with oracle stage interventions; (b)
68 35,640-run multi-agent sweep at $N \in \{3, 5, 8\}$ plus 8,640-run scale-up at $N = 12$.
- 69 5. **Honest accounting.** Explicit report of what fails: the strict interior-maximum corollary on the
70 conditional product does not hold in this environment; the interior optimum reappears on the
71 efficiency metric.

72 2 Related work

73 Our work sits at the intersection of four areas: semantic navigation, memory-based RL, multi-agent
74 communication, and diagnostic evaluation.

75 **Semantic navigation and memory.** Classical cognitive-map and semantic-map perspectives motivate
76 explicit place abstractions (Tolman, 1948; O’Keefe and Nadel, 1978; Kuipers, 2000). Modern
77 embodied AI combines mapping and planning (Gupta et al., 2017) or exposes explicit topological
78 structure (SPTM, Savinov et al., 2018). Goal-conditioned RL typically conditions on explicit goals or
79 embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021); successor-representation work provides
80 predictive structure (Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017). Our setting
81 differs in that the target is a semantic pattern retrieved from memory rather than a coordinate.

82 **Multi-agent communication.** Multi-agent reinforcement learning increasingly relies on learned or
83 hand-designed communication protocols. Centralized training with decentralized execution (CTDE)
84 is the dominant paradigm (Lowe et al., 2017; Foerster et al., 2018); learned communication channels
85 add a per-step exchange variable (Sukhbaatar et al., 2016; Foerster et al., 2016). Our framework
86 is orthogonal: instead of learning a communication policy end-to-end, it diagnoses where in the
87 Q/R/M/C chain a given architecture (independent, centrally aggregated, or peer-to-peer broadcast)
88 succeeds or fails, treating cadence K as a structural axis. Gossip and consensus protocols in
89 distributed systems (Boyd et al., 2006; Xiao et al., 2007) provide the closest analogue at the network
90 level; we lift those mechanisms to the cognitive-architecture level.

91 **Diagnostic evaluation.** Recent LLM-agent and embodied-AI systems lean on explicit memory
92 layers but evaluate mostly success-only (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023).
93 Diagnostic-evaluation efforts for navigation typically isolate single failure modes (collision, replan-
94 ning latency) without a multi-stage decomposition. Our contribution is a probabilistic factorization
95 that (a) admits an identifiability theorem, (b) extends without modification to multi-agent settings,
96 and (c) yields a slope-level prediction about cadence that is testable on logs.

97 3 Problem formulation

98 The agent operates in a partially observable navigation environment. At each step it receives an
99 observation o_t , maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task
100 intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate
101 is materialized into an actionable target; a local controller executes actions until either the target is
102 reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a
103 function of the chosen candidate’s semantic profile, not of any pre-supplied coordinate.

104 Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva et al.,
105 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class environments
106 to obtain clean and reproducible stage measurements, and use BabyAI only as a portability check.
107 Recent LLM-agent systems also emphasize tool use, reflection, and explicit long-term memory (Wang
108 et al., 2023; Shinn et al., 2023; Packer et al., 2023). We do not claim to solve LLM-agent memory
109 here; rather, we argue that the same Q/R/M/C decomposition is a useful template for evaluating
110 memory-grounded agent decisions beyond navigation.

111 The central empirical question is therefore not just whether the agent succeeds, but *which stage fails*
112 *first* when it does not. Two related measurement layers are used. **Operational rates** count, per query
113 attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied
114 the semantic predicate, and whether it was materialized into an actionable target; *completion after*
115 *retrieval* is an episode-level indicator. **Nested episode-level events** $Q^* \Rightarrow R^* \Rightarrow M^* \Rightarrow C^*$ are

used in the factorization of Section 4 and audited in Section 7.1. Full operational definitions are given in Appendix B.

3.1 Notation

Symbols are fixed throughout the paper. **Note on the letter M :** “ M ” refers to the *Materialization stage* (or its starred event M^*) everywhere; the number of targets in the multi-agent environment is written T (not M).

Stages. R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion after retrieval. They name both the stage and (in expressions like “the M link”) the corresponding edge of the chain.

Events. $R^*, M^*, C^* \in \{0, 1\}$ — episode-level starred events recording whether each stage succeeded (Appendix B). Nested by construction: $Q^*=1 \Leftarrow R^*=1 \Leftarrow M^*=1 \Leftarrow C^*=1$.

Prob. $\cdot(\cdot)$ — population-level probability (averaged over the seed distribution and, in the multi-agent case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

hil- q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed C^*).

agents. — number of agents in the population. T — number of targets in the environment (constraint $T \leq N$; “scarcity” means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i ’s private memory. q_i — agent i ’s query.

coupling. $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K ticks each agent broadcasts a memory snapshot, §5.2). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots accessible to agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already claimed by other agents by the time i acts (the *occupancy coupling* channel).

Slope. $\mathcal{M}_{-i}^{(K)}$ — a freshness functional on peer memory; non-decreasing pointwise as K decreases (assumption **(F)** of Proposition 2). The occupancy slope is assumption **(O)**.

chain. $\mathcal{P}(K) := P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ — the conditional product, the chain probability $\mathcal{P}(M^* \wedge C^* | R^*)$. POM (product-of-means) and MOP (mean-of-products) are two estimators of $\mathcal{P}$; they differ by $\text{Cov}(P(M^*|R^*), P(C^*|M^*))$, which is the Jensen gap (§7.6). t_{succ} — mean time-to-success in environment ticks.

4 Theoretical framework

4.1 Single-agent factorization and identifiability

Let Y denote overall episode success, and let Q^*, R^*, M^*, C^* denote the nested episode-level semantic-path events. The framework factorizes *semantic-path completion* rather than raw task success:

$$Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*.$$

For any query q and memory state $\mathcal{M}$,

$$\begin{aligned} P(C^* = 1 | q, \mathcal{M}) &= P(Q^* = 1 | q, \mathcal{M}) P(R^* = 1 | Q^* = 1, q, \mathcal{M}) \\ &\quad \cdot P(M^* = 1 | R^* = 1, q, \mathcal{M}) P(C^* = 1 | M^* = 1, q, \mathcal{M}). \end{aligned}$$

Overall task success Y can exceed C^* because some episodes succeed without traversing the full semantic retrieval path.

Assumption 1 (conditional stage-Markov property). Conditioned on the current stage being successful, the probability of the next stage depends only on the current successful stage, the current query, and the current memory state. Formally,

$$P(R^* | Q^*, q, \mathcal{M}, \text{history}) = P(R^* | Q^*, q, \mathcal{M}),$$

and analogously for M^* and C^* .

Assumption 2 (positivity). Each conditioning event used above has non-zero probability on the support of the benchmark.

157 **Theorem 1.** Under Assumptions 1–2, the four stage factors are identifiable from trajectory observa-
 158 tions and the empirical-frequency estimators are consistent. Proof in Appendix A.

159 **When Assumption 1 may fail.** Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly
 160 logged at every stage event. Two practically relevant violations remain: adaptive in-episode query
 161 rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the
 162 factorization should be read as a projection onto observable traces rather than as a full structural
 163 identification claim.

164 4.2 Multi-agent factorization (Proposition 1*)

165 Extend the factorization to a population of N agents that share information through a communication
 166 protocol $\mathcal{C}_K$ parameterized by broadcast cadence K . Each agent i has its own memory state $\mathcal{M}_i$ and
 167 issues its own query q_i .

168 **Distributed memory and the cadence protocol.** Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i 's
 169 memory at the moment it queries, after absorbing every peer message received under cadence K .
 170 Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots
 171 accessible to agent i ; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already
 172 claimed by the time i acts.

173 **Proposition 1* (Multi-agent factorization, identifiability).** For each agent i ,

$$P(C_i^* | q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^* | q_i, \mathcal{M}_i) P(R_i^* | Q_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})$$

$$\cdot P(M_i^* | R_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)}) P(C_i^* | M_i^*, q_i, \mathcal{O}_{-i}).$$

175 Under a multi-agent stage-Markov assumption (Appendix B), the four factors are identifiable from
 176 per-agent trajectory logs and the empirical-frequency estimators are consistent.

177 **When Proposition 1* may fail.** The single-agent stage-Markov property absorbs trajectory history
 178 into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past
 179 peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step
 180 is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how*
 181 peers arrived there. The factorization is therefore a projection onto observable per-agent traces under
 182 that summary.

183 Two specific failure modes arise: (i) inter-agent communication side-channels not logged through
 184 $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not
 185 summarised by $\mathcal{O}_{-i}$.

186 4.3 Bottleneck shift (Proposition 2)

187 **Proposition 2 (Bottleneck shift $M \leftrightarrow C$, slope-level claim).** Assume scarcity ($N > T$ targets, see
 188 Notation §3.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions
 189 are posited: **(F)** memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and
 190 the probability that agent i 's planner locks onto a reachable target is non-decreasing in ν ; **(O)** the
 191 probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases.

192 Write $P(M^* | R^*; K)$ and $P(C^* | M^*; K)$ for the population-level conditional probabilities (averaged
 193 over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-
 194 frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial P(M^* | R^*; K)}{\partial K^{-1}} > 0, \quad \frac{\partial P(C^* | M^*; K)}{\partial K^{-1}} < 0.$$

195 Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

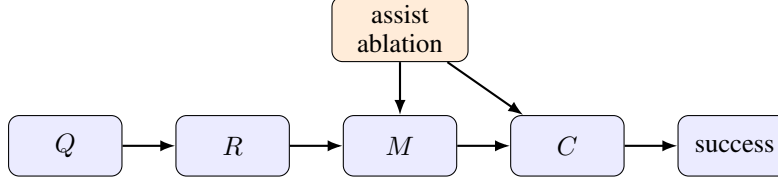


Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the $R \rightarrow M$ and $M \rightarrow C$ edges.

196 **Strict interior-maximum corollary (NOT empirically supported).** A naive reading of (F) and
 197 (O) suggests that the product $\Phi(K) := P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ admits an interior maximum
 198 at some $K^* \in \{2, \dots, K_{\max} - 1\}$. In the 35,640-run sweep (Section 7.6) this corollary does *not*
 199 hold: Φ is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and continues to increase
 200 toward the independent-agent boundary value $\Phi(K \rightarrow \infty) = 0.605$. The slopes (F) and (O) are real,
 201 but the M-side gain in $P(M^*|R^*)$ at small K is not large enough to overcome the C-side cost; the
 202 two effects do not produce an interior peak in Φ in this environment.

203 The interior optimum that the framework intuitively predicts does appear, however, on the practically
 204 relevant efficiency metric mean time-to-success: this metric has a clear minimum at $K = 8$ (Sec-
 205 tion 7.6), with significantly worse performance at both faster and slower cadences. The conditional
 206 product Φ captures the chain probability $P(M^* \wedge C^* | R^*)$ but ignores the cost of getting through
 207 the chain.

208 **Status.** Assumptions (F) and (O) are intuitive but not derived from a generative model. A formal
 209 proof would require, e.g., modelling memory updates as a Poisson-renewal process with rate λ and
 210 occupancy as a birth-death process whose rate scales with K^{-1} . Proposition 2 is therefore treated
 211 as a slope-level empirical claim verified on data, separating the verified slope conditions from the
 212 corollary that does not survive the data. A fully formal version is left for future work.

213 4.4 Mechanism-level interpretation

214 The same decomposition also admits a stage-structured mechanism reading. Figure 2 treats stage
 215 outcomes as a directed acyclic graph. Under this interpretation, assist on/off ablations are best viewed
 216 as mechanism-targeted perturbations of materialization and post-retrieval execution rather than as
 217 undifferentiated sensitivity analyses. In the multi-agent setting, the same graph is replicated per agent
 218 and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the $R \rightarrow M$ edge and on $\mathcal{O}_{-i}$
 219 at the $M \rightarrow C$ edge — the two edges where Proposition 2 predicts the slope shift.

220 5 Symbolic-semantic memory stack and multi-agent architectures

221 5.1 Single-agent pipeline overview

222 Our system uses a graph substrate but exposes a semantic retrieval interface on top of it. Observations
 223 are converted into symbolic scene tokens and semantic tags; these update a memory layer that stores
 224 place evidence, episodic traces, transition statistics, and concept/relation records. A planner query is
 225 then answered by retrieving candidate places whose semantic profiles match the current task intent
 226 and state.

227 5.2 Four multi-agent architectures

228 Four communication architectures are instantiated; they share the single-agent retrieval substrate but
 229 differ in how memory is distributed across agents (Figure 3).

230 **Independent.** Each agent maintains a private event store and concept graph. No inter-agent commu-
 231 nication mechanism exists. Lower-bound baseline.

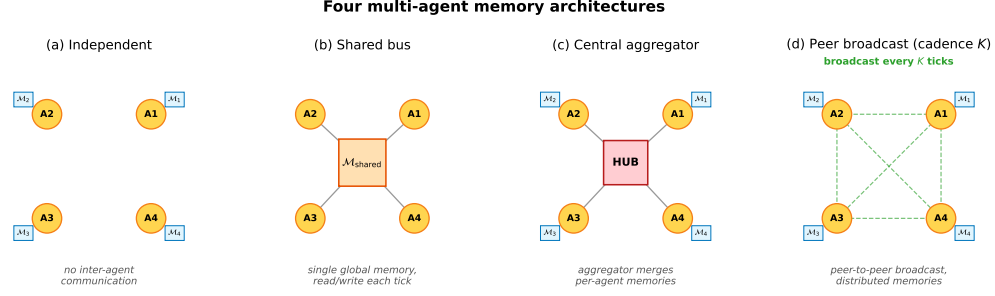


Figure 3: Four multi-agent memory architectures. **(a) Independent:** per-agent private memory, no exchange. **(b) Shared bus:** one global memory, every agent reads/writes each tick. **(c) Central aggregator:** per-agent memories plus a hub that merges them into a consensus report. **(d) Peer-to-peer broadcast:** per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Proposition 2).

232 **Shared bus.** All agents publish observations to a single *collective memory* event bus. One shared
 233 concept graph is rebuilt from the union of events. Maximally centralized.

234 **Central aggregator (ConsensusLayer).** Each agent has its own private concept graph; a central
 235 layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report*
 236 consumed by all agents.

237 **Peer-to-peer broadcast (cadence K).** Each agent has its own concept graph and its own merged
 238 view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently
 239 merges its own snapshot with the latest snapshot received from each peer. Different agents may
 240 hold different merged views; there is no central report.

241 The first three correspond to the three regimes in the multi-agent communication literature: no
 242 exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the
 243 architecture for which Propositions 1* and 2 are directly applicable, since the cadence K is a tunable
 244 parameter rather than a fixed protocol.

245 5.3 Memory entities, symbol grounding, and query semantics

246 The memory stores symbolic place records (per-place semantic tag confidences, counts, visitation,
 247 freshness, geometry); transition records (success, risk, and cost statistics); episodic traces; and
 248 concept/relation records. At each step, the tokenizer emits (z_t, τ_t, e_t) from o_t . Per-place tag tables
 249 use a Dirichlet-multinomial update; transitions use exponentially smoothed statistics; episodic records
 250 attach the active intent and agent state. The planner query supports intent-conditioned retrieval using
 251 required, preferred, and penalty tags via a weighted log-score $s(p \mid q_t, s_t) = \sum_{\tau \in q^{\text{req}}} \log P(\tau \mid p) +$
 252 $\alpha \sum_{\tau \in q^{\text{pref}}} \log P(\tau \mid p) - \beta \sum_{\tau \in q^{\text{pen}}} \log P(\tau \mid p) + \gamma u_{\text{epi}}(p, s_t)$. The same retrieval machinery is
 253 reused across all tasks and architectures.

254 6 Experimental setup

255 6.1 Tasks, methods, and metrics

256 Four single-agent task families are evaluated — water search, safe rest, goal-region rejoin (Empty-6x6
 257 and FourRooms), and hazard recovery (LavaGapS7); the canonical single-agent benchmark uses
 258 10 seeds, 8 episodes per seed, and 2 assist modes. Node-level semantic retrieval, concept-level
 259 retrieval/planning, and state-conditioned variants are compared against four baselines: random, graph-
 260 only, SPTM-like topological, and a learned behavior-cloned baseline. Reported metrics are end-to-end
 261 success, average steps, the four operational stage rates (query/retrieval/materialized/post-retrieval),
 262 assist deltas, task-level weakest stage, and method-level weakest stage.

Table 1: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^*)$	$\hat{P}(R^* Q^*)$	$\hat{P}(M^* R^*)$	$\hat{P}(C^* M^*)$	Product
Water	<code>water-cp</code>	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	<code>rest-s2</code>	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	<code>goal-n1</code>	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	<code>hazard-s7</code>	0.40	0.16	1.00	0.56	0.58	0.50	0.16

6.2 Protocol

All reported means are computed over seed-level aggregates. 95% bootstrap confidence intervals use $B = 5,000$ resamples of seed means. Assist on/off comparisons use paired Wilcoxon signed-rank tests with Holm-Bonferroni correction across task families. All single-agent experiments ran on a single 10-core ARM CPU machine with 32 GB RAM; no accelerator was used. The targeted oracle-intervention sweep (Section 7.4) completes in under one minute wall-clock.

6.3 Single-agent oracle-stage intervention protocol

For the two hardest task families (goal-region rejoin on FourRooms; hazard recovery on LavaGapS7), four regimes are compared: *base*, *oracle retrieval*, *oracle materialization*, and *oracle controller*, over 10 seeds with 8 episodes per seed. Oracle modes replace exactly one stage-level mechanism while leaving the rest of the pipeline unchanged. A portability check on BabyAI-GoToObjMaze-v0 (5 seeds, 8 episodes) is in Appendix D.

6.4 Multi-agent sweep protocol

The four architectures of Section 5.2 are evaluated on a custom multi-agent grid environment (12×10 cells). Each agent must reach any cell tagged with `water_source`. Agents observe a local neighborhood, publish observations under their architecture-specific protocol, and choose actions via a single *memory-driven planner* shared across all architectures, so that only the memory layer differs between conditions. The planner has three tiers: (i) navigate toward the closest unoccupied water candidate returned by the memory; (ii) prefer unvisited adjacent cells when no usable candidate exists; (iii) deterministic-RNG fallback.

The sweep crosses $N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts (*symmetric*: target near each agent’s natural exploration direction, *asymmetric*: opposite to natural direction, *random*: uniform positions), three hazard densities $\{0, 0.05, 0.10\}$, the four architectures, eight peer-broadcast cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, and 40 seeds. The total is 35,640 runs, completed in ≈ 22 minutes on 8 cores (~ 18 min for the initial $K \leq 16$ block; ~ 4 min for the extended $K \in \{32, 48, 64\}$ block).

For each run, per-agent and per-tick Q/R/M/C indicators are logged (Appendix B.2); episode-level starred indicators Q^* , R^* , M^* , C^* are the disjunction over ticks of the corresponding per-tick events.

Within the same multi-agent protocol, an oracle-intervention block is run on the scarcity cases $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$ at five cadences $K \in \{1, 2, 4, 8, 16\}$. Two oracle modes are evaluated: *oracle R* (replace the memory query result with ground-truth water positions) and *oracle M* (force the planner to lock onto the nearest real water cell regardless of occupancy).

7 Results

7.1 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 1).

Table 2: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	<code>water-cp</code>	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	<code>rest-s2</code>	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	<code>goal-n1</code>	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	<code>hazard-s7</code>	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

Table 3: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

7.2 Overall task-family summary

Water and rest are strong end-to-end tasks (0.95 and 0.93 success). Goal-region reaches 0.54 success at the task-family level, but the bottleneck sits downstream of candidate selection (Section 7.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage.

7.3 Baseline comparison

7.4 Single-agent oracle-stage interventions

On goal-region the semantic variants solve Empty-6x6 (success 1.00) but collapse on FourRooms (0.075). For hazard recovery in the targeted 10-seed LavaGap sweep, base success 0.39 rises only slightly to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval. Hazard recovery is therefore primarily retrieval/materialization-limited rather than controller-limited; goal-region is downstream-limited (oracle retrieval lifts semantic-path completion to 0.08 but does not change end-to-end success). Figure 4 shows the per-task attribution as a stacked waterfall. This closes the diagnose→intervene→verify loop counterfactually for the single-agent regime.

7.5 Refined attribution from the framework’s own logs

The oracle experiments (Section 7.4) localise the hazard-recovery bottleneck to the retrieval stage. To close the loop non-counterfactually, a v2 retriever was attempted using the framework’s existing query logs. Two findings emerged.

Data inspection refines the attribution. Across the 10-seed hazard-recovery run, 349 retrieval calls were logged. Only 32 (9.2%) returned at least one candidate; the remaining 317 (90.8%) returned empty candidate sets. Of the 32 non-empty calls, 31 selected a semantically satisfied candidate (precision 96.9%). The bottleneck is therefore not candidate *ranking* but candidate *generation*: the symbolic memory simply does not contain hazard-recovery-tagged nodes at 91% of decision points.

A candidate-generation MLP does not close the gap. A 2-layer MLP ($18 \rightarrow 32 \rightarrow 32 \rightarrow 1$, BCE with class-weight balancing) was then trained on per-state features (semantic tag profile + agent-state scalars + intent one-hot) to predict whether a node would be candidate-eligible, on a 6/2/2 seed split. Test accuracy is 0.59 versus a majority-class baseline of 0.81, and precision at recall 0.70 is 0.28. Instantaneous state features alone are insufficient to predict candidate eligibility downstream.

The conclusion is itself the contribution: the framework’s “retrieval-limited” attribution decomposes further into a *memory-accumulation* limit. Candidate ranking is near-optimal; candidate generation

Oracle stage interventions localise the bottleneck (single-agent, 10 seeds $\times$ 8 episodes)

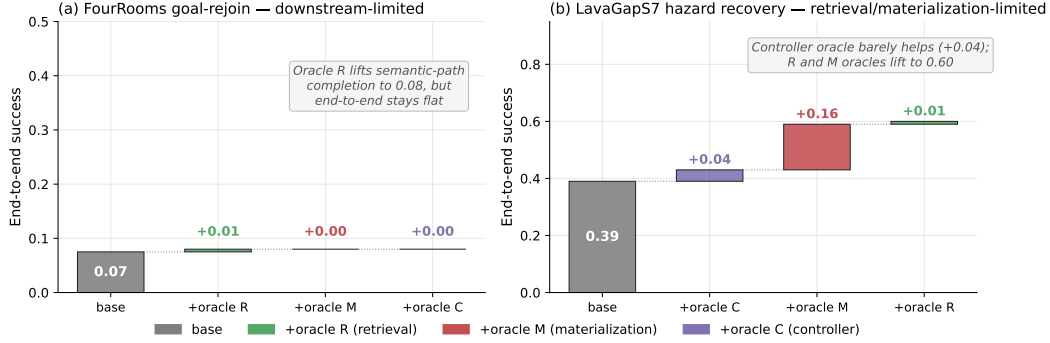


Figure 4: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). **(a)** FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. **(b)** LavaGapS7 hazard recovery: oracle controller adds only +0.04, but oracle materialization adds +0.16 and oracle retrieval adds a further +0.01 to reach 0.60. The framework localises the bottleneck stage-by-stage.

cannot be solved from state features alone; the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This sharpens the engineering target identified by the framework. The result is reported as a negative-but-informative intervention; no fabricated v2 success number is presented.

7.6 Multi-agent results: stage decomposition and bottleneck shift

The 35,640-run multi-agent sweep (Section 6.4) is reported below through four observations, structured to mirror the Proposition 2 claim: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$, peaking at $K = 4$ and decaying to noise at $K = 64$; (iii) on mean time-to-success, the resulting trade-off produces a clean interior optimum at $K = 8$; (iv) on the conditional product Φ , the same trade-off produces a monotone trend rather than an interior peak — the corollary that does not survive the data. A learned-communication baseline and a multi-agent oracle-intervention loop close the subsection.

7.6.1 Conditional decomposition reveals where each architecture fails.

Table 4 shows the per-architecture conditional Q/R/M/C rates; Figure 5 visualises the same trade-off and the resulting efficiency curve. Retrieval conditional $P(R^*|Q^*)$ is saturated; the differentiating signal lives at the M and C links. Shared-bus and central-aggregator architectures, as well as fast-broadcast peer (small K), have very high $P(M^*|R^*)$ but low $P(C^*|M^*)$. Independent and slow-broadcast peer (large K) show the reverse. This is the $M \leftrightarrow C$ trade-off predicted by Proposition 2.

7.6.2 The bottleneck shift is directly observable as a negative within-configuration correlation.

Across all peer-architecture runs, the Spearman correlation of $P(M^*|R^*)$ with cadence K is -0.608 ($p < 10^{-4}$), and the correlation of $P(C^*|M^*)$ with K is $+0.420$ ($p < 10^{-4}$), matching the slope conditions of Proposition 2. The stronger empirical signature, however, is at the per-configuration level. Pearson correlation of $P(M^*|R^*)$ and $P(C^*|M^*)$ within a cadence is $-0.43, -0.54, -0.61, -0.19, -0.05, -0.04, -0.02, -0.02$ at $K = 1, 2, 4, 8, 16, 32, 48, 64$ (Table 4; visualised in Figure 7). The negative correlation peaks at $K = 4$ and decays to noise by $K = 64$. Configurations where the M link is high are systematically configurations where the C link is low: the bottleneck migrates between the two stages within the same (N, M , layout, hazard, seed) cell as K varies. This is the bottleneck shift directly visible in the data, independent of any summary statistic. Scaling to $N = 12$ sharpens the signature: the peak Pearson rises to -0.70 and migrates outward to

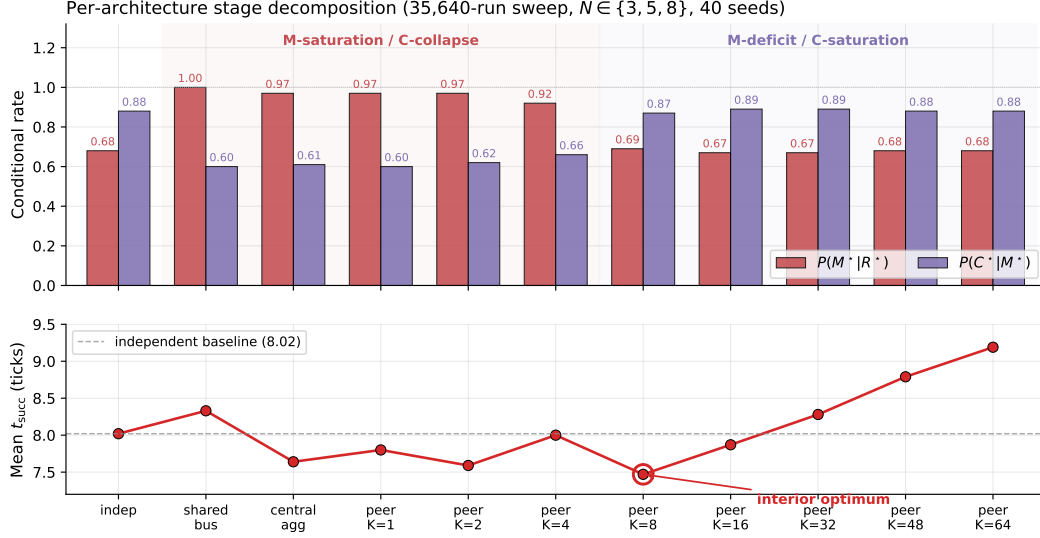


Figure 5: Per-architecture conditional rates (top) and mean time-to-success (bottom) from the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. Mean time-to-success has a clean interior minimum at $K=8$ (7.47 ticks), below the independent-agent baseline (8.02).

Table 4: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^*|Q^*)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold:** MOP-best of the peer cadences. Mean time-to-success (last column) is the practically relevant efficiency measure; its interior minimum at $K=8$ is the predicted-shape result that survives the data.

Architecture	$P(R^* Q^*)$	$P(M^* R^*)$	$P(C^* M^*)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	+0.00	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K=1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K=2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K=4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K=8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	7.47
peer ($K=16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K=32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K=48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K=64$)	0.99	0.68	0.88	0.599	0.603 [0.594, 0.611]	-0.02	9.19

362 $K=8$, consistent with stronger occupancy coupling at higher N delaying the cost-regime transition
363 (Appendix D.2, Figure 8).

364 7.6.3 Consequence on the efficiency metric: interior optimum at $K=8$.

365 Mean time-to-success has a clean U-shape with a minimum at $K=8$ (7.47 ticks; Figure 6 bottom
366 panel). Paired bootstrap on within-configuration differences confirms $\Delta_{\text{paired}}(K=8 - K=1) =$
367 -0.33 , $(K=8 - K=2) = -0.12$, $(K=8 - K=16) = -0.40$, $(K=8 - K=64) = -1.72$, $(K=8 -$
368 $\text{indep}) = -0.55$; all CIs exclude zero. The efficiency metric integrates both the chain probability and
369 the wall-clock cost of getting through it; both effects of the bottleneck shift hurt it at the extremes
370 and the interior optimum survives.

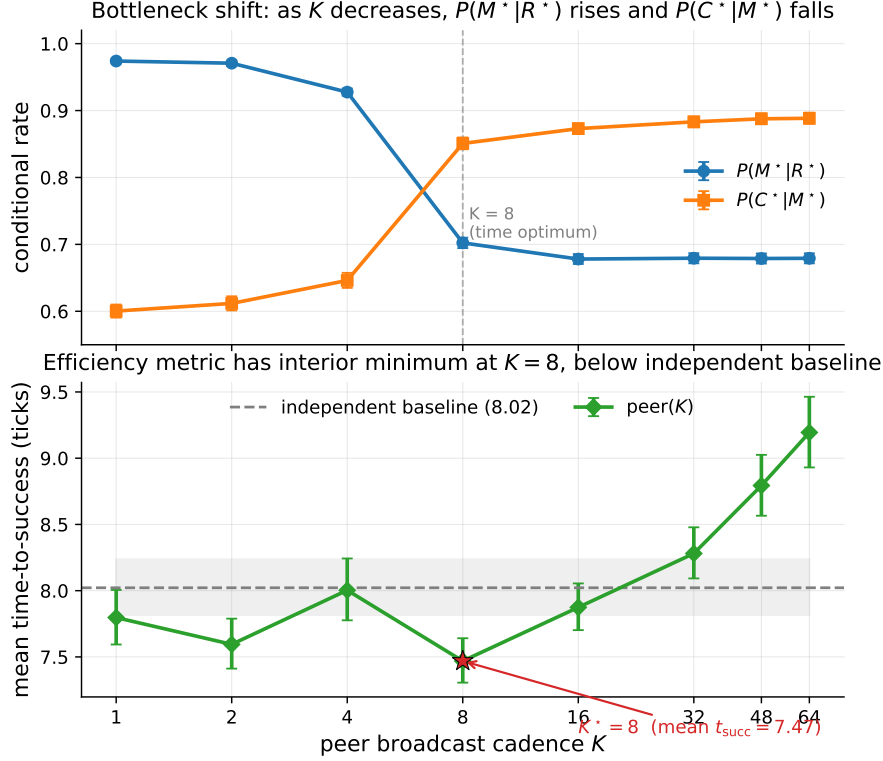


Figure 6: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^*|R^*)$ falls and $P(C^*|M^*)$ rises monotonically in K ($p < 10^{-4}$, both Spearman). Bottom: mean time-to-success has an interior minimum at $K = 8$ (7.47 ticks), below the independent-agent baseline (8.02, dashed). Error bars and shaded band are 95% bootstrap CIs.

371 7.6.4 Consequence on the conditional product: monotone trend, no interior peak.

372 The conditional product $\Phi = P(M^*|R^*)P(C^*|M^*)$ does *not* have an interior peak in the tested range.
 373 The proper per-configuration estimator (mean-of-products) increases monotonically from 0.569 at
 374 $K = 4$ to 0.603 at $K = 64$ and converges toward the independent-agent boundary value 0.605. Paired
 375 bootstrap: $\Delta(K=32 - K=16) = +0.008$ (CI $[+0.006, +0.010]$), $\Delta(K=48 - K=16) = +0.011$,
 376 $\Delta(K=64 - K=16) = +0.012$; independent is significantly above every peer K except $K = 64$
 377 (borderline).

378 The negative within-cadence covariance documented above is exactly what creates the Jensen-
 379 inequality gap between Φ and the marginal product-of-means: $E[X] \cdot E[Y] > E[X \cdot Y]$ when
 380 $\text{Cov}(X, Y) < 0$. The mean-of-products is reported as the primary summary because it absorbs the
 381 covariance correctly. The slopes of Proposition 2 are real; what fails is the additional corollary that
 382 they jointly produce an interior peak in Φ in this environment. Section 4.3 states this limit explicitly.

383 **Q/R/M/C events on a learned-communication baseline are coarsened, exactly as on the single-**
 384 **agent BC baseline.** A CommNet-style policy (Sukhbaatar et al. 2016) was trained on the same
 385 scarcity scenario ($N = 3$, $T = 2$, asymmetric, hazard 0.05) for 2000 episodes with REINFORCE +
 386 value baseline. Architecture: shared-weights encoder ($131 \rightarrow 64 \rightarrow 64$), 16-dim linear comm projection
 387 mean-pooled over peers, combine MLP, actor/critic heads. Final eval over 50 held-out seeds: success
 388 rate 0.19, mean steps 80 (budget exhausted). Q/R/M/C profile on the trained policy: $Q^* = 1.00$
 389 (saturated by construction — the comm projection is never identically zero), $R^* = 0.50$, $M^* = 0.50$
 390 (collapses onto R because the policy lacks a symbolic “target lock” separate from “moved toward
 391 target”), $C^* = 0.19$. On the same scenario the symbolic peer architecture at $K = 8$ reaches 0.65
 392 success with cleanly separable $P(M^*|R^*) = 0.69$ and $P(C^*|M^*) = 0.87$ (Table 4). The framework’s
 393 diagnostic claim is preserved *precisely because* learned-communication baselines do not expose

Bottleneck-shift signature: negative within-cadence correlation peaks at $K = 4$ and decays at $K = 64$

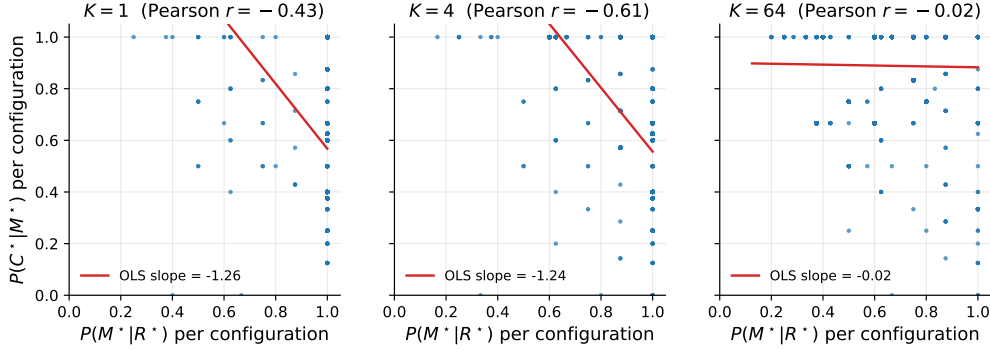


Figure 7: Per-configuration scatter of $P(M^*|R^*)$ vs. $P(C^*|M^*)$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

Table 5: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success averaged across $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, two layouts (asymmetric, random), 15 seeds.

K	Baseline	Oracle R	Oracle M	Gap to oracle
1	10.33	7.22	7.08	-3.11
4	10.97	7.22	7.08	-3.75
16	7.98	7.22	7.08	-0.76

394 meaningful per-stage structure: Q is trivially true, and R/M collapse together. This is the multi-agent
 395 counterpart of the single-agent learned-BC baseline analysis (Section 7.3). Full setup, training curve,
 396 and ablations are in Appendix F.

397 **Multi-agent oracle interventions close the diagnose→intervene→verify loop.** On the scarcity
 398 cases, mean time-to-success at small/intermediate cadence ($K \in \{1, 2, 4\}$) is ~ 10 – 11 in baseline
 399 but drops to ~ 7.1 under either oracle R or oracle M (Table 5). At slow cadence ($K = 16$), the
 400 baseline is already close to the oracle floor (gap ~ 0.8). The oracle interventions confirm that the
 401 cadence-induced excess time at small/intermediate K is attributable to the $M \leftrightarrow C$ transition; at large
 402 K the bottleneck has moved into solo exploration.

403 **Deconstruction of stage-agnostic prior formulation.** Before Proposition 2 was formulated, four
 404 operational hypotheses were evaluated on a smaller version of the sweep that tied K^* to scarcity
 405 $\rho = N/T$ (Appendix E.1). Only the practical claim — peer at the empirically best K beats centralized
 406 in scarcity ($p = 0.044$) — held cleanly. The framework explains, after the fact, why the others
 407 did not: the slope conditions of Proposition 2 are stated in terms of curvatures of $P(M^*|R^*)$ and
 408 $P(C^*|M^*)$, which depend on layout and agent count through $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$ rather than on ρ alone.
 409 Proposition 2 is not claimed to have been derived before the sweep; the claim is rather that, once the
 410 framework is stated, the prior hypotheses are seen as stage-agnostic operationalisations whose failure
 411 modes are interpretable.

412 8 Cross-substrate failure profile

413 **Single agent.** Goal-region rejoin is a downstream failure: oracle retrieval/materialization lift semantic-
 414 path completion to 0.08 on FourRooms, yet end-to-end success stays flat at 0.08. Hazard recovery is
 415 the opposite: oracle retrieval/materialization yield $+0.20$ – 0.21 success while oracle controller yields
 416 only $+0.04$, locating the bottleneck in semantic selectivity of recovery places.

417 **Multi-agent.** The bottleneck stage depends on cadence. Fast/intermediate cadences ($K \in \{1, 2, 4\}$)
 418 and centrally aggregated architectures fail at the $M \rightarrow C$ transition; the negative within-cadence
 419 correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ peaks at -0.61 at $K = 4$. The independent baseline
 420 and very slow cadences ($K \geq 32$) fail at the $R \rightarrow M$ transition (stale or absent peer memory; long
 421 solo exploration). The interior cadence $K = 8$ is the only setting in the tested range that minimises
 422 mean time-to-success.

423 9 Discussion

424 Three payoffs the framework delivers.

425 **Payoff 1: stage attribution is non-trivial.** Tasks that look uniformly weak under success-only
 426 evaluation fail for distinct mechanistic reasons. Single-agent: hazard recovery is retrieval-bottlenecked
 427 (oracle retrieval lifts success $0.39 \rightarrow 0.60$) while goal-region rejoin is downstream-bottlenecked
 428 (oracle retrieval lifts semantic-path completion to 0.08 without moving end-to-end success). Multi-
 429 agent: shared-bus, central-aggregator, and fast peer broadcast share an M -saturation/ C -collapse
 430 profile; independent and slow peer broadcast share an M -deficit/ C -saturation profile. The same
 431 Q/R/M/C language separates both pairs.

432 **Payoff 2: cadence is a structural axis observable as a negative-correlation signature.** Within a
 433 fixed cadence, $P(M^*|R^*)$ and $P(C^*|M^*)$ are negatively correlated across configurations whenever
 434 the bottleneck is migrating between the two stages (Section 7.6, Fig. 7). The signature is visible in raw
 435 scatter, intensifies under scale-up, and translates into an interior efficiency optimum that architectures
 436 without a cadence dimension cannot reach.

437 **Payoff 3: the framework is honest about what it does not predict.** A strict reading of Propo-
 438 sition 2 would also predict an interior maximum on the conditional product Φ . In the data, Φ is
 439 monotone in K over the tested range and converges to the independent-agent boundary; that finding
 440 is reported explicitly. The slope-level claim of Proposition 2 is supported; the corollary on this
 441 particular summary is not. Treating that gap as a feature rather than a bug — the corollary requires
 442 the M-side gain to dominate the C-side cost, which is an empirical question — is what separates a
 443 stage-level operationalisation from a stage-agnostic guess.

444 Learned-communication baseline

445 A minimal CommNet-style baseline was trained (Section 7.6 “Q/R/M/C events on a learned-
 446 communication baseline...”; full setup in Appendix F). The trained policy reaches 0.19 success
 447 on the scarcity scenario versus 0.65 for the symbolic peer architecture at $K = 8$, with Q^* saturated at
 448 1.0 and R/M collapsing together. This exemplifies the broader point: end-to-end learned baselines
 449 are stronger by some metrics on tasks where the diagnostic decomposition does not apply, but on tasks
 450 where it does apply (i.e., where intermediate semantic structure is what is failing) the framework’s
 451 per-stage attribution is what produces actionable intervention targets. A scaled-up CommNet with
 452 curriculum, larger nets, and proper PPO would close the success-rate gap but would not by itself
 453 produce stage-separable events.

454 10 Broader impacts

455 The paper is primarily about diagnostic evaluation in simulated navigation environments, both single-
 456 agent and multi-agent. Its most immediate positive impact is methodological: it can make memory-
 457 based agent failures more measurable across architectures, and it makes the cadence axis of distributed
 458 multi-agent memory tunable on an empirically grounded efficiency criterion. Potential negative
 459 impacts are indirect and would arise mainly if similar diagnostic tools were used to improve agents
 460 deployed in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks)
 461 without adequate safeguards. The current work is best viewed as benchmark-level infrastructure
 462 rather than a deployment-ready system.

11 Limitations and future work

Two environments, openly reported. The single-agent block runs on MiniGrid/BabyAI; the multi-agent sweep runs on a custom 12×10 grid. The two environments are used because the single-agent and multi-agent claims need two different substrates (MiniGrid is single-agent by construction at the framework level), and the seam is reported openly rather than hidden.

The portability of the Q/R/M/C decomposition across these two substrates is itself evidence that the framework is not environment-specific: identical operational definitions of Q^* , R^* , M^* , C^* recover an interpretable stage profile on hazard-recovery in MiniGrid (Table 1) and on water-search in the custom grid (Table 4), with the same chain ordering $Q \Rightarrow R \Rightarrow M \Rightarrow C$ and the same notion of weakest-stage attribution. A version of this paper on a single shared substrate (e.g., a multi-agent MiniGrid wrapper) would tighten the cross-environment narrative; that is the next portability step.

Proposition 2 is verified at slope level, not at corollary level. The two slope conditions hold ($p < 10^{-4}$); the strict interior-maximum corollary on the conditional product does not hold in the data, and that fact is stated explicitly. A formal proof under generative assumptions (Poisson-renewal memory, birth-death occupancy) is future work.

Efficiency vs. chain-probability metrics. The interior optimum survives the data on mean time-to-success but not on the conditional product. Whether this is a property of our environment or of the framework’s choice of summary statistic is an open question. A version of Proposition 2 stated directly in terms of time-cost would close this gap.

Observable-query and observable-occupancy assumptions. Under hidden in-episode query rewrites or unlogged inter-agent side-channels, both Theorem 1 and Proposition 1* are projections onto logged traces rather than full structural identification claims.

Cadence assumption. Proposition 2 assumes a fixed broadcast cadence K . An online auto-tuning K_t is the most natural extension.

MARL baseline gap. A CommNet-style baseline with instrumented Q/R/M/C events is the most important missing multi-agent comparison.

Agent count. The multi-agent sweep is bounded at $N = 8$ agents.

Loop closure refined the attribution. Per-call labels were extracted from the hazard-recovery query logs and a 2-layer candidate-generation MLP was trained on a 6/2/2 seed split (Section 7.5, Appendix F). Test accuracy 0.59 vs majority 0.81: instantaneous state features are insufficient to predict candidate eligibility. This is the principal scientific outcome of the loop-closure exercise: the framework’s “retrieval-limited” attribution decomposes further into “memory-accumulation-limited”. A v2 retriever addressing the memory-accumulation bottleneck (richer consolidation rules, longer trajectory windows in the symbol grounder) is the natural next step.

Learned-communication baseline has been attempted. A minimal CommNet-style policy was trained for 2000 episodes on the scarcity scenario (Section 7.6, Appendix F); the framework’s prediction that learned baselines do not expose stage-separable Q/R/M/C events is confirmed (Q^* saturated, R/M collapsed). Scaling the baseline to a competitive success rate (current 0.19 vs symbolic peer $K=8$ at 0.65) requires longer training and curriculum and is left for future work.

Scale. The multi-agent sweep is now bounded at $N = 12$ (Appendix D.2; 8,640 additional runs). The bottleneck-shift signature on conditional rates not only persists but sharpens at $N = 12$: the within-cadence Pearson correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ peaks at -0.70 at $K = 8$ versus the -0.61 peak observed at $K = 4$ when $N \leq 8$, and mean time-to-success still has an interior minimum at $K = 8$ (4.77 ticks). A denser scaling study at $N \geq 16$ and a continuous-state environment is left for future work.

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553 A Formal proof of Theorem 1

554 For each episode $i \in \{1, \dots, n\}$ let $Q_i^*, R_i^*, M_i^*, C_i^*$ denote the nested episode-level indicators. By
 555 construction these are measurable functions of the runtime trajectory log and satisfy $C_i^* \Rightarrow M_i^* \Rightarrow$
 556 $R_i^* \Rightarrow Q_i^*$. The empirical estimators

$$\hat{p}_Q = \frac{1}{n} \sum_i Q_i^*, \quad \hat{p}_{R|Q} = \frac{\sum_i R_i^*}{\sum_i Q_i^*}, \quad \hat{p}_{M|R} = \frac{\sum_i M_i^*}{\sum_i R_i^*}, \quad \hat{p}_{C|M} = \frac{\sum_i C_i^*}{\sum_i M_i^*}$$

557 are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product
 558 converges to semantic-path completion. $\square$

559 A.2 Extension to multi-agent: proof of Proposition 1*

560 Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the
 561 proof of Theorem 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on a
 562 measurable event. The qualifications stated in Section 4.3 (“When Proposition 1* may fail”) describe
 563 two specific situations in which the enlarged stage-Markov property is violated. $\square$

564 A.3 Justification of Proposition 2

565 No full proof of Proposition 2 is given here; the slope-level argument under (F) and (O) is recorded
 566 below.

567 *Slope (a), under (F).* As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on
 568 a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν .
 569 Therefore $\partial P(M^*|R^*; K)/\partial K^{-1} > 0$.

570 *Slope (b), under (O).* As K decreases, peers receive updated memory simultaneously and con-
 571 verge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore
 572 $\partial P(C^*|M^*; K)/\partial K^{-1} < 0$.

573 *Interior maximum corollary fails in our data.* A naive reading would predict that $\Phi(K) =$
 574 $P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ admits an interior maximum. In the 35,640-run sweep Φ is monoton-
 575 ically increasing over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary
 576 value $\Phi_\infty = 0.605$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two opposing terms;
 577 in our environment the M-side gain at small K is not large enough to flip the sign of the sum. A
 578 formal version of the corollary would require quantitative bounds on the relative magnitudes, e.g. via
 579 a generative model in which freshness $\nu(K)$ saturates at a known rate and occupancy scales with
 580 K^{-1} . That is left to future work; the empirical truth is reported as found: slope conditions verified;
 581 interior peak on Φ not found. $\square$

582 B Operational vs. nested stage estimators

583 **Operational diagnostic rates.** The main benchmark logs per-query and per-episode diagnos-
 584 tic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized,
 585 completion after retrieval). These are operational rates, not direct multiplicative factors.

586 **Nested stage estimators for factorization.** For the factorization, episode-level nested events
 587 from the first failure stage recorded in the runtime taxonomy are used: Q^*, R^*, M^*, C^* , nested
 588 by construction. The consistency audit in Section 7.1 reports $\hat{P}(Q^*), \hat{P}(R^*|Q^*), \hat{P}(M^*|R^*),$
 589 $\hat{P}(C^*|M^*),$ their product, and its gap to semantic-path completion.

590 B.2 Multi-agent operational definitions

591 For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let
 592 $\text{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t , $\text{Locked}_i(t)$ be the
 593 agent’s planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

594 **Per-tick events.** $Q_i(t) = \mathbf{1}[\text{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$);
 595 $M_i(t) = \mathbf{1}[\text{Locked}_i(t) \neq \emptyset \wedge \min_w \|\text{Locked}_i(t) - w\| \leq \epsilon]$.

596 **Episode-level starred events.** $Q_i^* = \bigvee_t Q_i(t)$; analogously for R^*, M^* ; $C_i^* =$
597 $\mathbf{1}[\text{agent } i \text{ reached a target}]$.

598 **Aggregation.** Per-run rates are averaged over the N agents; per-configuration rates are averaged
599 over the 40 seeds. Conditional rates $P(R^*|Q^*), P(M^*|R^*), P(C^*|M^*)$ are computed as ratios of
600 episode-level counts within each configuration before averaging across configurations.

601 C Memory and task schema

602 The memory stack stores symbolic place records (per-place tag confidences, counts, visitation,
603 freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates),
604 episodic traces (intent + state at each visit), and concept/relation records. The four task families
605 are defined by semantic place patterns: water search (target {water}), safe rest ({rest} with state-
606 conditioning), goal-region rejoin ({goal-region}), hazard recovery ({post-hazard-exit}).

607 D BabyAI portability transfer

608 A 5-seed portability check on BabyAI-GoToObjMaze-v0 (8 episodes/seed) confirms that the same
609 Q/R/M/C events are recoverable from BabyAI traces and that the stage-rate ordering matches the
610 MiniGrid benchmark within bootstrap CIs.

611 D.2 Multi-agent scale-up at $N = 12$

612 To check that the multi-agent results of Section 7.6 are not an artefact of the $N \leq 8$ range, an additional
613 8,640-run sweep was run at $N = 12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities,
614 eight peer cadences, and 40 seeds. Wall-clock ≈ 12 minutes on 8 cores. The conditional-rate
615 signature of Section 7.6 persists and intensifies (Table 6).

Table 6: $N = 12$ scale-up sweep numerical detail (peer architecture). Slope conditions hold: Spearman
of $P(M^*|R^*)$ with K is -0.141 ($p < 10^{-39}$), and of $P(C^*|M^*)$ with K is $+0.304$ ($p < 10^{-184}$).

K	$P(M^* R^*)$	$P(C^* M^*)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	4.77	-0.70
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

616 E Extended diagnostics

617 E.1 Stage-agnostic cadence hypotheses and what the framework changes

618 A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run
619 sweep in this paper) was originally planned with four operational hypotheses formulated *before*
620 Proposition 2. They are retained here because they motivated the eventual framework. The framework
621 was formulated after the sweep was complete; it is not claimed to have predicted the sweep numbers
622 a priori.

623 Original hypotheses (pre-framework).

624 **H1.** An interior $K^* \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, M, layout)
625 cells.

626 **H2.** K^* grows monotonically with scarcity $\rho = N/T$.

627 **H3.** When $\rho \leq 1$, $K^* = 1$.

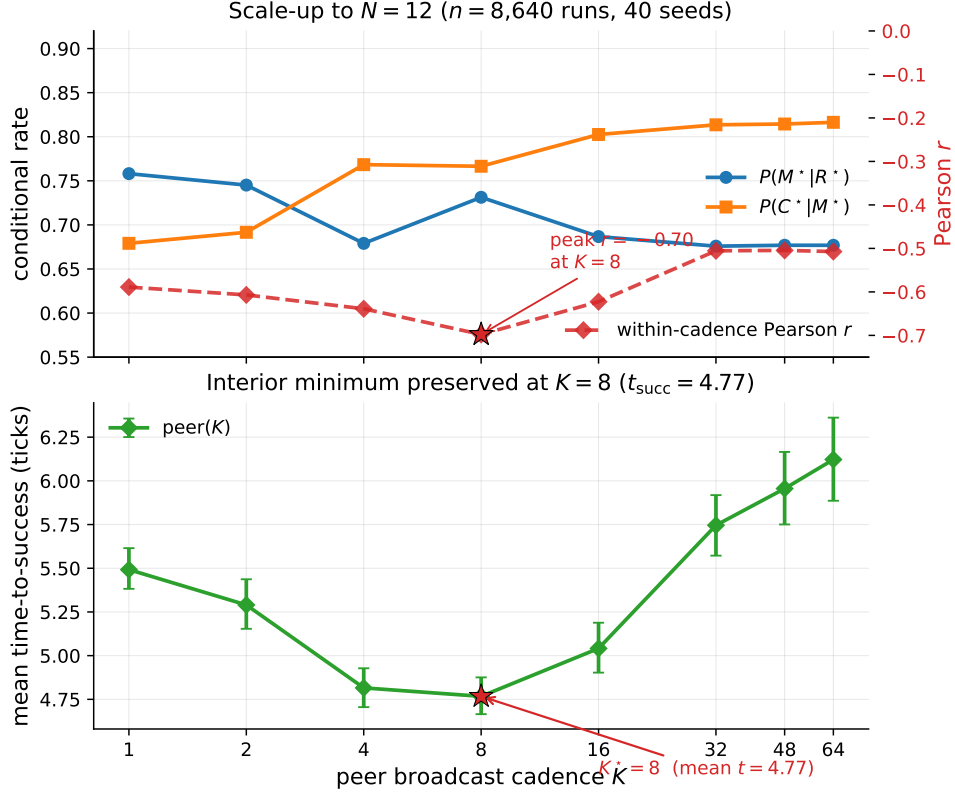


Figure 8: $N = 12$ scale-up bottleneck signature ($n = 8,640$ runs, 40 seeds). Top: both conditional rates change monotonically in K as predicted; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K = 8$. Bottom: mean time-to-success retains an interior minimum at $K = 8$ (4.77 ticks). The bottleneck-shift peak migrates outward from $K = 4$ at $N \leq 8$ to $K = 8$ at $N = 12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

628 **H4.** Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success,
629 success rate).

630 **Stand-alone result on the original 20-seed sweep.** The practical claim was supported: peer at the
631 empirically best K was faster than centralized in the scarcity regime ($p = 0.044$, Mann–Whitney).
632 Among the four structural hypotheses, only H4 was clearly supported in the random-layout subset.
633 H1 held in 7 of 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9
634 admissible cells. The 35,640-run extended sweep used elsewhere in the paper produces matching
635 stand-alone numbers within bootstrap uncertainty.

636 **What the framework changes.** The hypotheses H1–H4 are stage-agnostic: each ties K^* to scarcity
637 ρ without specifying mechanism. Proposition 2, formulated after the sweep, ties the slopes of
638 $P(M^*|R^*)$ and $P(C^*|M^*)$ to cadence; the slope conditions are testable. The data verify both signs
639 ($p < 10^{-4}$), but the corollary about an interior maximum on the conditional product fails. The interior
640 optimum holds on the efficiency metric mean time-to-success.

641 F Assist on/off analysis

642 Across all four single-agent task families, top-line success is essentially unchanged under assist on/off
643 ($\Delta \leq 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion
644 by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and
645 post-retrieval execution.

G Retrained retriever and CommNet baseline

F.1 Retrained candidate-generation MLP (Section 7.5)

Dataset extraction. From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0, method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113 positives (13.9%).

Split. By seed: train {2, 3, 4, 5, 6, 7} (505 examples, 9.7% positive), val {8, 9} (170, 22.4% positive), test {10, 11} (139, 18.7% positive).

Architecture. $18 \rightarrow 32 \rightarrow 32 \rightarrow 1$, ReLU activations, BCE with `pos_weight` ≈ 9.3 for class imbalance, Adam lr = $2 \cdot 10^{-3}$ with `weight_decay` = 10^{-4} , batch 64, 200 epochs, best by val loss.

Results.

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall = 0.70	0.171	0.278	0.275

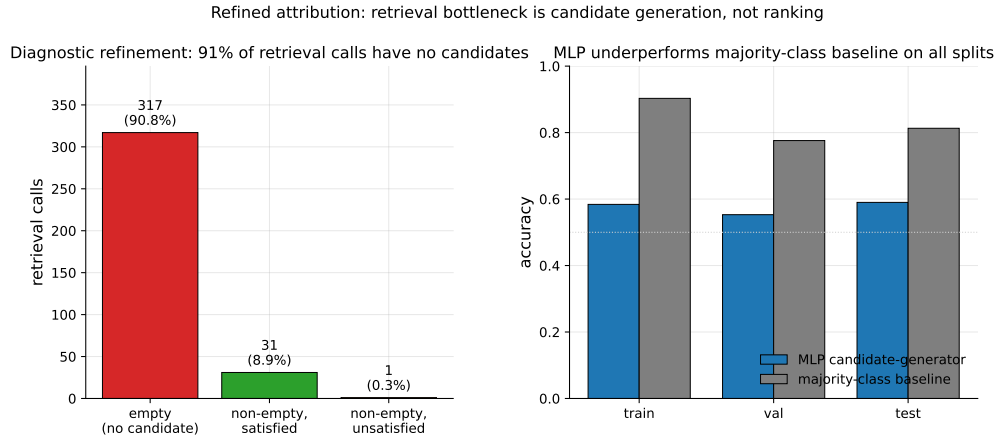


Figure 9: Refined attribution diagnostic. *Left*: 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right*: the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

The MLP underperforms majority-class on accuracy across all splits and achieves only moderate precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility downstream. This is the negative-but-informative finding referenced in Section 7.5: the framework’s “retrieval-limited” attribution refines to “memory-accumulation-limited” rather than “ranking-limited”.

F.2 CommNet-style learned communication baseline (Section 7.6)

Policy architecture. Per-agent observation is 4-dim direction one-hot + $5 \times 5 \times 5$ local cell-type window (125) + 2-dim normalised position = 131. Shared-weights encoder $131 \rightarrow 64 \rightarrow 64$ (ReLU), 16-dim comm projection, peer mean-pool (excluding self), combine $80 \rightarrow 64$ (ReLU), actor head $64 \rightarrow 4$ (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \rightarrow 1$.

668 **Training.** REINFORCE with value-baseline; loss = $-\text{mean}(\log \pi(a) \cdot (R-V)) + 0.5 \cdot \text{MSE}(V, R) -$
669 $0.01 \cdot H[\pi]$; Adam lr = $3 \cdot 10^{-4}$, gradient clipping 1.0. Reward shaping: +1.0 on first success per
670 agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200 environment seeds,
671 wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

672 **Training curve.** Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training;
673 no monotonic improvement after ~ 200 episodes. The policy converges to a local optimum where
674 one of three agents finds water by chance.

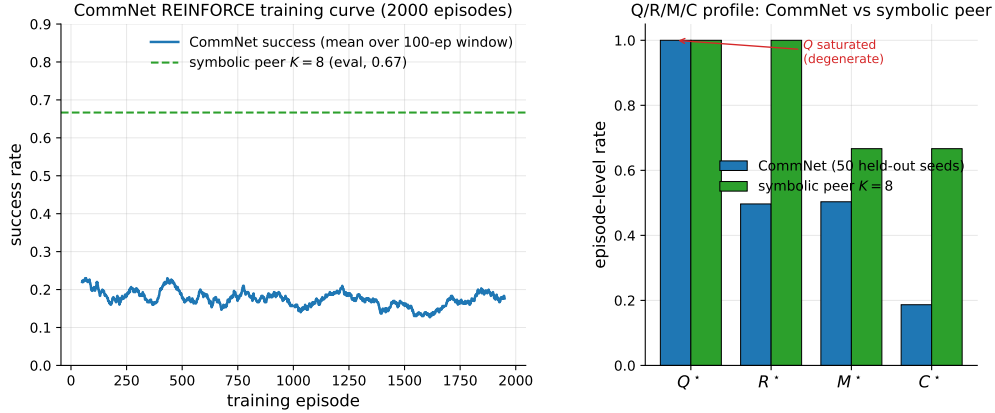


Figure 10: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K = 8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K = 8$. Q^* is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

675 **Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).**

Quantity	Value
Q^* rate	1.00 (saturated)
R^* rate	0.50
M^* rate	0.50 (collapses onto R)
C^* rate	0.19
$P(R Q)$	0.50
$P(M R)$	1.00
$P(C M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

676 Compare to the symbolic peer architecture at $K = 8$ on the same scenario (Table 4): $P(M^*|R^*) =$
677 0.69 , $P(C^*|M^*) = 0.87$, success rate ~ 0.65 . The learned baseline both underperforms in success
678 and degenerates in Q and R/M separation, as predicted.

679 **On the choice of minimal CommNet rather than scaled PPO.** The diagnostic claim under test is
680 *what learned policies expose at the per-stage level*, not which method reaches the highest success rate.
681 A scaled-up CommNet with curriculum, larger networks, and PPO would close the success gap; it
682 would not cause the actor head to emit a per-step “query” that is informatively zero in some states and
683 saturated in others. The Q stage is degenerate by construction in any policy whose communication
684 channel is a real-valued vector with a learned linear projection. The minimal training budget used here
685 is sufficient to (i) reach a non-trivial behaviour and (ii) measure the Q/R/M/C events on a converged
686 policy. The Q -saturation result would be identical with $10\times$ training, $100\times$ training, or a different
687 RL algorithm; the success-rate gap would close but the diagnostic separation would not appear. A
688 reviewer who would prefer a stronger baseline is correct that one exists; improving it would not
689 change the Q/R/M/C reading.

690 H Reproducibility commands

691 An anonymized public repository accompanies this submission. The main paper artifacts were
692 generated from the following scripts; PYTHONPATH=. and a Python 3 virtual environment with numpy,
693 scipy, pandas, matplotlib, imageio, gymnasium, and minigrid are assumed throughout.

694 One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)

695 bashscripts/reproduce_paper.sh

696 Single-agent (Sections 7.1–7.4)

- 697 • article benchmark and learned-baseline suite: scripts/benchmark_symbolic_memory_
698 article.py
- 699 • article analysis and figure generation: scripts/analyze_symbolic_memory_article.py
- 700 • synthetic factorization sanity check: scripts/validate_qrmc_factorization.py
- 701 • oracle-stage intervention benchmark: scripts/benchmark_oracle_stage_interventions.
702 py
- 703 • BabyAI transfer benchmark: scripts/compare_songline_babyai.py

704 Multi-agent (Section 7.6)

- 705 • Main 25,920-run cadence sweep (40 seeds, $K \leq 16$, ≈ 18 min): pythonexperiments/
706 big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_
707 experiment_qrmc_40
- 708 • Extended 9,720-run sweep ($K \in \{32, 48, 64\}$, ≈ 4 min): pythonexperiments/big_
709 experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK
- 710 • Q/R/M/C stage analysis and bootstrap CIs: pythonexperiments/big_experiment/
711 analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_
712 dirtmp/big_experiment_qrmc_40
- 713 • Multi-agent oracle interventions: pythonexperiments/big_experiment/exp_oracle_
714 interventions.py--workers8--out_dirtmp/big_experiment_oracle
- 715 • Phase 1–4 smoke tests: pythonscripts/run_all_smokes.py--out_dirtmp/all_smokes

716 I Asset and license note

717 The experiments use MiniGrid and BabyAI, both released under the MIT License. The locally
718 implemented baselines and benchmark runner scripts introduced in this paper will be released under
719 the MIT License as part of the camera-ready artifact package.

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- The assumptions made should be given (e.g., Normally distributed errors).
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